Specific heat

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use mass, material and temperature change to calculate heating energy.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 21.*
15 J other; fraction 0.70 or 70%.

02. Worked example: explain the model
$2 \times 100 \times 3 = 600$ J; positive heat input raises temperature under assumptions.

03. Read and interpret the model
A; larger c means smaller ΔT for fixed Q and m.

04. Apply the physics
ΔT 5°C; Q 1000 J.

05. Explain your reasoning
Energy for warming depends on the temperature change, not the arbitrary zero of the temperature scale.

06. Review: connect an earlier idea *Review: Unit 4, Day 12.*
24 J, by conservation of mechanical energy.

07. Exit check
600 J.

08. STRETCH: test the model
6°C and 3°C; doubling mass halves the rise at fixed Q and c.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 21.*
Initial U=150 J; K=144 J; v≈9.798 m/s.

02. Worked example: explain the model
2×100×3=600 J; positive heat input raises temperature under assumptions.

03. Read and interpret the model
A; larger c means smaller ΔT for fixed Q and m.

04. Apply the physics
ΔT 30°C; final 48°C.

05. Explain your reasoning
Energy for warming depends on the temperature change, not the arbitrary zero of the temperature scale.

06. Review: connect an earlier idea *Review: Unit 4, Day 12.*
No. That height requires an increase mg×2 in potential energy, exceeding the initial energy; it would require negative final K.

07. Exit check
2100 J/(kg·°C); no phase change and energy input reaches sample without unaccounted losses.

08. STRETCH: test the model
c=300 J/(kg·°C); Q=5400 J, assuming constant c and no phase change.